MILITARY SPECIFICATION

ADHESIVE: GRINDING DISK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of grinding disk adhesive (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-637	Steel Bar, Carbon, Cold Finished (Standard Quality, Free Machining)
PPP-C-96	Cans, Metal 28 Gage and Lighter
PPP-D-729	Drums, Shipping and Storage, Steel, 55 Gallon
PPP-P-704	Pail, Metal (Shipping, Steel, 1 through 12 Gallon)

STANDARDS

FEDERAL

Fed. Test Method Std. No. 175	Adhesives: Methods of Testing
Fed. Std. No. 313	Material Safety Data Sheets, Preparation and Submission of

/FSC 8040/

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-LS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM D 903 Test for Peel or Stripping Strength of Adhesive Bonds
ASTM D 1084 Tests for Viscosity of Adhesives
ASTM G 23 Recommended Practice for Operating Light and Water Exposure Apparatus (Carbon-Arc Type) for Exposure of Nonmetallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATIONS, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic, Tariff Order Section, 1616 P Street N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. Latitude will be allowed in the selection of raw materials, provided the finished product complies with all the requirements of this specification.

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3.2 Brushing Properties. The grinding disk adhesive, as received, shall be of brushing consistency (see 4.4).

3.3 Odor. The dried film of grinding disk adhesive shall be free of obnoxious or objectionable odor (see 4.5.1).

3.4 Viscosity. The grinding disk adhesive shall have a viscosity of not less than 0.6 nor more than 0.9 Pa·s (600-900 cP) when determined as specified in 4.5.2.

3.5 Peel Strength. The peel strength of the grinding disk adhesive shall be 1750 N/m (10 lbf/in) minimum at 23 ± 1 C (73 ± 2 F), 1580 N/m (9 lbf/in) minimum at 23 ± 1 C (73 ± 2 F) after accelerated aging, and 1050 N/m (6 lbf/in) minimum at 65 ± 1 C (149 ± 2 F) when determined as specified in 4.5.3.

3.6 Total Solids. The total solids in the grinding disk adhesive shall be 34 to 40 percent when determined as specified in 4.5.4.

3.7 Workmanship. The finished adhesive shall be homogeneous and free of foreign matter.

3.8 Material Safety Data Sheet. When specified by the procuring activity, (see 6.2) a Material Safety Data Sheet shall be prepared and submitted in accordance with Fed. Std. No. 313.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.3 Sampling.

4.3.1 For Examination of Adhesive. Sampling of unit containers for examination of adhesive shall be conducted in accordance with level S-1 of MIL-STD-105.

4.3.2 For Examination of Packaging, Packing and Marking. A random sample of unit containers shall be selected from each lot in accordance with MIL-STD-105 for examination of packaging, packing and marking. Unless otherwise specified, the inspection level shall be S-4.

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4.4 Examination. Sample units selected in accordance with 4.3.1 and 4.3.2 shall be examined for defects at the acceptable quality levels (AQL) shown in table I.

TABLE I. Classification of defects

Item	AQL percent	Classification of defects (critical)	Defect (none defined)	Method of inspection
Adhesive (see 3.2, 3.3, 3.7 & 4.3.1)	2.5	Major 101	Not of brushing consistency	Visual
		Major 102	Objectionable odor	Olfactory
		Major 103	Not homogeneous	Visual
		Major 104	Contains foreign matter	Visual
Unit container (see 4.3.2 & 5.1)	2.5	Major 105	Improper type	Visual
		Major 106	1/ Improper size	2/ Visual
		Major 107	Improper fill	Approved scale
		Major 108	Leakage	Visual
		Major 109	Improper closure	Visual
		Major 110	Improper marking	Visual
Box open (see 4.3.2 & 5.2)	2.5	Major 111	Improper type	Visual
		Major 112	Lack of, or improper strapping	Visual
Box closed (see 4.3.2, 5.2 & 5.3)	2.5	Major 113	Gross weight (max)	Approved scale
		Major 114	Improperly closed	2/ Visual
		Major 115	Improper marking	Visual

1/ The actual weight of a container filled with the minimum required quantity of adhesive shall be the basis for determining the acceptable weight of subsequent containers.

2/ Approved by the procuring activity.

4.5 Tests.

4.5.1 Odor. The odor of the adhesive shall be determined in accordance with Fed. Test Method Std. No. 175, Method 4051.

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4.5.2 Viscosity. The viscosity of the adhesive shall be determined in accordance with ASTM D 1084, Method B.

4.5.3 Peel Strength.

4.5.3.1 Specimens. Thirty panels of steel 25 mm x 102 mm (1 in x 4 in) and conforming to QQ-S-637, composition 1020, shall be prepared by sanding and cleaning with a suitable, completely volatile, organic solvent in accordance with the manufacturer's recommendations. Thirty strips 25 mm x 305 mm (1 in x 12 in) of number 20 grit abrasive cloth shall be bonded to the steel panels in accordance with the manufacturer's direction and in accordance with ASTM D 903.

4.5.3.2 Procedure. The specimens prepared as specified in 4.5.2.1 shall be tested as follows: One third of the specimens shall be tested in accordance with ASTM D 903. One third of the specimens shall be tested in accordance with ASTM D 903 after the accelerated aging at 50 ± 2 C (122 ± 4 F) for 7 days (168 hours) in accordance with ASTM G 23. The remaining one third shall be tested in accordance with ASTM D 903 except that the testing temperature shall be 65 ± 1 C (149 ± 2 F).

4.5.4 Total Solids. Pour approximately 10 grams of a representative sample of the adhesive into a low-form weighing bottle, cover and weigh. After removing the cover, apply heat at 70 ± 2 C (158 ± 4 F) until the sample reaches constant weight. The sample shall be cooled in a desiccator prior to weighing. Calculate the percentage of total solids as follows:

$$\text{Total solids, percent} = \frac{\text{Weight of residue} \times 100}{\text{Weight of sample}}$$

CAUTION: Care should be exercised when volatile materials are of such a nature as to constitute a toxic, fire or explosive hazard.

5. PACKAGING

5.1 Packaging. Grinding disk adhesive shall be packaged according to level A or C, as specified (see 6.2).

5.1.1 Level A. Grinding disk adhesive shall be furnished in 473 milliliter (pint), 946 milliliter (quart), 3785 milliliter (gallon) cans, in 18.9 liter (5 gallon) pails or in 208 liter (55 gallon) drums, as specified (see 6.2).

5.1.1.1 Cans. The 473 milliliter (pint), 946 milliliter (quart) and 3785 milliliter (gallon) cans shall conform to PPP-C-96, Type V, Class 2. Exterior coating shall be at the option of the contractor. The 3785 milliliter (gallon) containers shall be provided with wire handles, galvanized or protectively coated to resist corrosion.

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5.1.1.2 Pails. The 18.9 liter (5 gallon) pails shall conform to Type II of PPP-P-704. Exterior coating shall be required and wire handles on pails shall be galvanized or protectively coated to resist corrosion.

5.1.1.3 Drums. The 208 liter (55 gallon) drums shall conform to Type III or Type IV of PPP-D-729.

5.1.2 Level C. Grinding disk adhesive shall be furnished in 473 milliliter (pint), 946 millilter (quart) or 3785 milliliter (gallon) cans, in 18.9 liter (5 gallon) pails or 208 liter (55 gallon) drums, in accordance with the manufacturer's commercial practice. The packaged quantities shall be specified (see 6.2).

5.2 Packing. Grinding disk adhesive shall be packed according to level A, B or C as specified (see 6.2).

5.2.1 Levels A and B. The adhesive packaged in accordance with 5.1 of like description and of capacities up to and including 3785 milliliter (gallon) cans shall be prepared for delivery in accordance with the overseas or domestic shipment requirements for levels A and B, respectively, in accordance with the Appendix to PPP-C-96. The adhesive furnished in 18.9 liter (5 gallon) pails and 208 liter (55 gallon) drums require no overpacking.

5.2.2 Level C. The adhesive packaged in accordance with 5.1 shall be packed in accordance with supplier's standard commercial practice in such a manner as to insure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional Marking. In addition to marking specified by 5.3, each interior package and exterior shipping container shall be marked with the date of manufacture of the adhesive, by month and year, not by code.

6. NOTES

6.1 Intended Use. The adhesive procured under this specification is intended for use by the bonding of abrasive disks to metal disks.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Quantity required.

c. When a Material Safety Data Sheet is required (see 3.8).

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d. Size of containers (see 5.1.1 or 5.1.2)

e. Selection of applicable level of packaging and level of packing required (see 5.1 and 5.2).

f. Whether special marking is required (see 5.3).

Custodians:

Army -- MR
Air Force -- 99

Review activities:

Air Force -- 85

User activities:

Army -- ME, ER, MI, CR

Preparing Activity:

Army -- MR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-14064C Adhesive: Grinding Disk (FSC 8040)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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